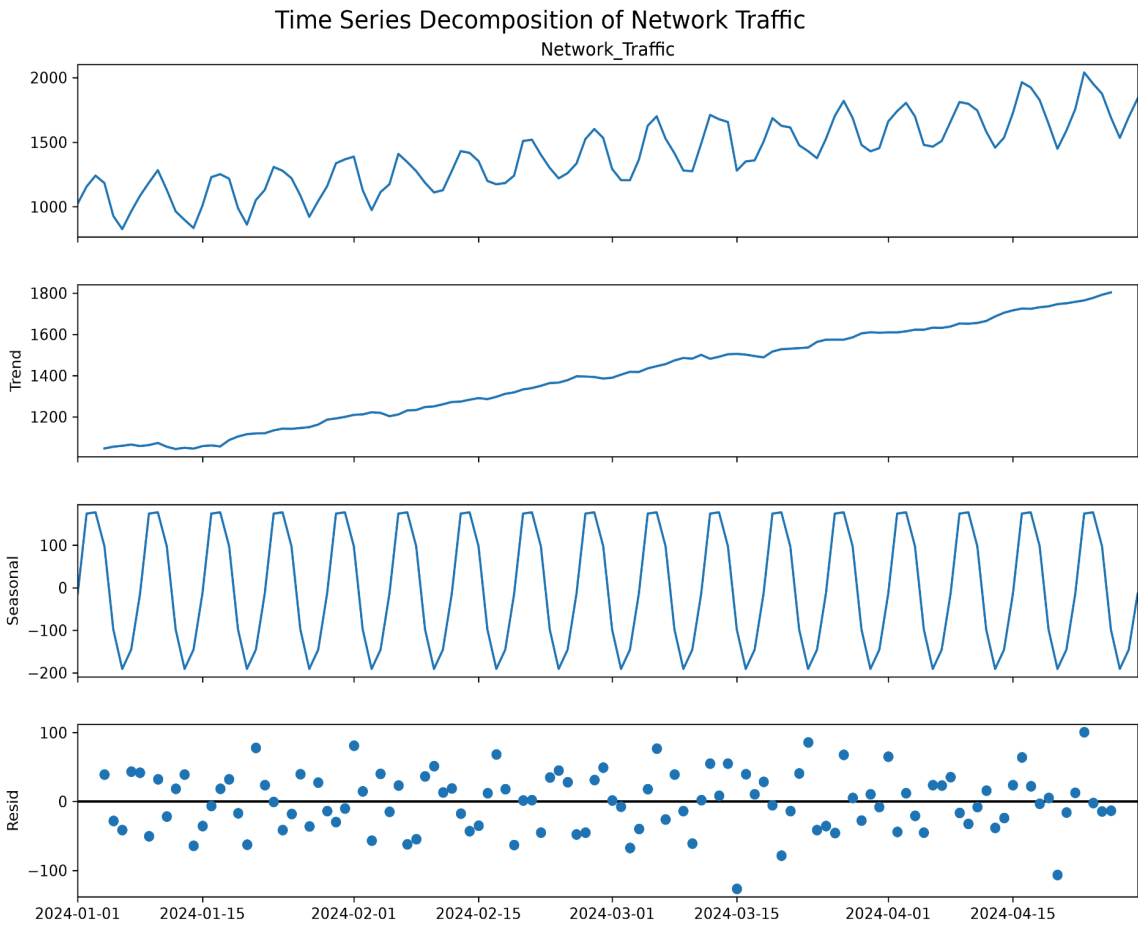
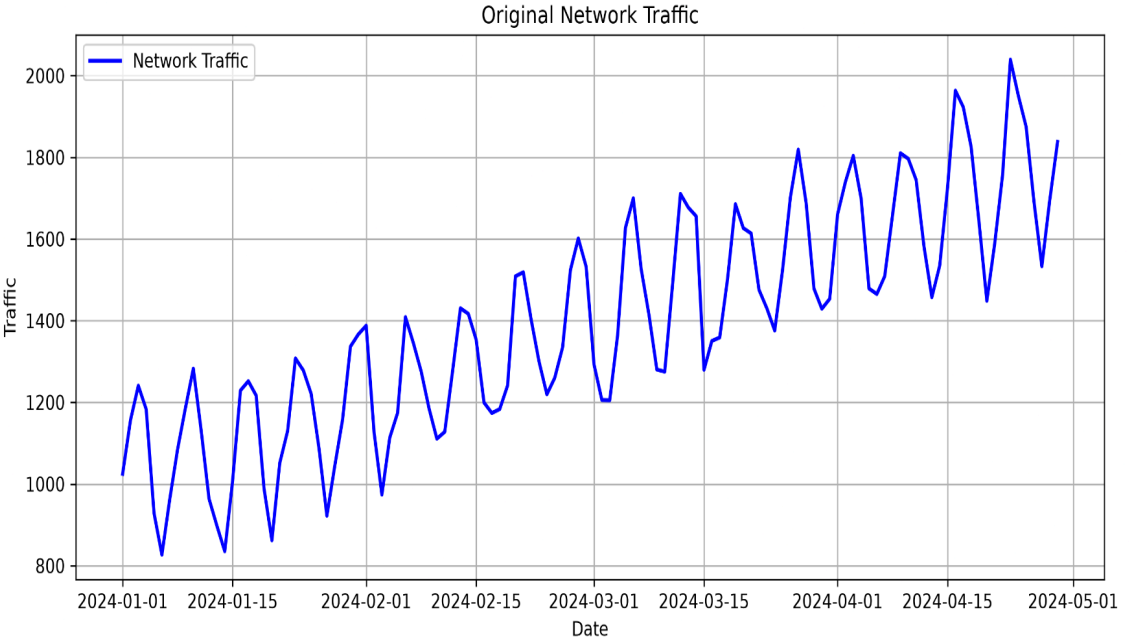


Experiment 1 output :



Experiment 2 output :

```
➤ /home/devan/meow/python/venv/bin/python  
/home/devan/meow/python/MCA.cpp/mlcs_exp_2.py
```

===== DATASET INFORMATION =====

Columns found:

```
['timestamp', 'packet_count']
```

First 5 records:

	timestamp	packet_count
0	01-01-26	1200
1	02-01-26	1300
2	03-01-26	1250
3	04-01-26	1400
4	05-01-26	1500

Time column selected : timestamp

Traffic column selected: packet_count

/home/devan/meow/python/MCA.cpp/mlcs_exp_2.py:55: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
data[time_column] = pd.to_datetime(
```

Number of observations: 35

Traffic statistics:

count	35.000000
mean	1702.571429
std	477.723418
min	1200.000000
25%	1425.000000
50%	1600.000000
75%	1800.000000
max	3500.000000

Name: packet_count, dtype: float64

===== DDoS THRESHOLD =====

Average traffic : 1702.57

Standard deviation: 477.72

DDoS threshold : 3135.74

===== ARIMA MODEL =====

```

ARIMA order: (5, 1, 2)
/home/devan/meow/python/venv/lib/python3.14/site-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: A date index has been provided, but it has no associated frequency information and so will be ignored when e.g. forecasting.
  self._init_dates(dates, freq)
/home/devan/meow/python/venv/lib/python3.14/site-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: A date index has been provided, but it has no associated frequency information and so will be ignored when e.g. forecasting.
  self._init_dates(dates, freq)
/home/devan/meow/python/venv/lib/python3.14/site-packages/statsmodels/tsa/base/tsa_model.py:473: ValueWarning: A date index has been provided, but it has no associated frequency information and so will be ignored when e.g. forecasting.
  self._init_dates(dates, freq)
/home/devan/meow/python/venv/lib/python3.14/site-packages/statsmodels/tsa/statespace/sarimax.py:966: UserWarning: Non-stationary starting autoregressive parameters found. Using zeros as starting parameters.
  warn('Non-stationary starting autoregressive parameters'
/home/devan/meow/python/venv/lib/python3.14/site-packages/statsmodels/tsa/statespace/sarimax.py:978: UserWarning: Non-invertible starting MA parameters found. Using zeros as starting parameters.
  warn('Non-invertible starting MA parameters found.')
/home/devan/meow/python/venv/lib/python3.14/site-packages/statsmodels/base/model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to ")

```

ARIMA model successfully trained.

Model Summary:

SARIMAX Results

```

=====
=====
Dep. Variable:    packet_count  No. Observations:      35
Model:           ARIMA(5, 1, 2)  Log Likelihood      -254.076
Date:            Tue, 15 Sep 2026  AIC                  524.152
Time:            17:57:57  BIC                      536.363
Sample:          0  HQIC                      528.316
                  - 35
Covariance Type:    opg
=====
=====

```

	coef	std err	z	P> z	[0.025	0.975]
ar.L1	0.0847	1.522	0.056	0.956	-2.899	3.068
ar.L2	0.4147	0.292	1.418	0.156	-0.158	0.988
ar.L3	-0.3165	0.769	-0.412	0.681	-1.823	1.190
ar.L4	-0.3865	0.421	-0.919	0.358	-1.211	0.438

```

ar.L5      -0.0511   0.576  -0.089   0.929  -1.180   1.078
ma.L1      -1.0707   1.536  -0.697   0.486  -4.081   1.940
ma.L2       0.3052   1.315   0.232   0.817  -2.273   2.883
sigma2     1.652e+05 8.28e+04 1.995   0.046 2876.907 3.27e+05
=====
=====
Ljung-Box (L1) (Q):          0.02  Jarque-Bera (JB):          0.67
Prob(Q):                   0.89  Prob(JB):                   0.72
Heteroskedasticity (H):      3.75  Skew:                   0.32
Prob(H) (two-sided):        0.04  Kurtosis:                2.74
=====
=====

```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

===== FORECAST =====

/home/devan/meow/python/venv/lib/python3.14/site-packages/statsmodels/tsa/base/tsa_model.py:837: ValueWarning: No supported index is available. Prediction results will be given with an integer index beginning at `start`.

```
    return get_prediction_index(
```

/home/devan/meow/python/venv/lib/python3.14/site-packages/statsmodels/tsa/base/tsa_model.py:837: FutureWarning: No supported index is available. In the next version, calling this method in a model without a supported index will result in an exception.

```
    return get_prediction_index(
```

Timestamp	Predicted_Traffic
2026-12-02	1821.848853
2026-12-03	1782.028202
2026-12-04	1797.633503
2026-12-05	1649.076234
2026-12-06	1545.398015
2026-12-07	1471.562002
2026-12-08	1465.326362
2026-12-09	1523.599995
2026-12-10	1596.975687
2026-12-11	1663.165268
2026-12-12	1686.944817
2026-12-13	1670.987626
2026-12-14	1627.216806
2026-12-15	1580.036517
2026-12-16	1550.364401
2026-12-17	1547.087253
2026-12-18	1567.165612
2026-12-19	1597.367470
2026-12-20	1623.168133
2026-12-21	1634.308059
2026-12-22	1628.802272

2026-12-23	1612.093159
2026-12-24	1593.354665
2026-12-25	1580.956340
2026-12-26	1578.980963
2026-12-27	1586.340454
2026-12-28	1598.163769
2026-12-29	1608.591678
2026-12-30	1613.446487
2026-12-31	1611.697669

===== DDoS PREDICTION =====

2026-12-02 00:00:00	Predicted Traffic: 1821.85	Status: NORMAL
2026-12-03 00:00:00	Predicted Traffic: 1782.03	Status: NORMAL
2026-12-04 00:00:00	Predicted Traffic: 1797.63	Status: NORMAL
2026-12-05 00:00:00	Predicted Traffic: 1649.08	Status: NORMAL
2026-12-06 00:00:00	Predicted Traffic: 1545.40	Status: NORMAL
2026-12-07 00:00:00	Predicted Traffic: 1471.56	Status: NORMAL
2026-12-08 00:00:00	Predicted Traffic: 1465.33	Status: NORMAL
2026-12-09 00:00:00	Predicted Traffic: 1523.60	Status: NORMAL
2026-12-10 00:00:00	Predicted Traffic: 1596.98	Status: NORMAL
2026-12-11 00:00:00	Predicted Traffic: 1663.17	Status: NORMAL
2026-12-12 00:00:00	Predicted Traffic: 1686.94	Status: NORMAL
2026-12-13 00:00:00	Predicted Traffic: 1670.99	Status: NORMAL
2026-12-14 00:00:00	Predicted Traffic: 1627.22	Status: NORMAL
2026-12-15 00:00:00	Predicted Traffic: 1580.04	Status: NORMAL
2026-12-16 00:00:00	Predicted Traffic: 1550.36	Status: NORMAL
2026-12-17 00:00:00	Predicted Traffic: 1547.09	Status: NORMAL
2026-12-18 00:00:00	Predicted Traffic: 1567.17	Status: NORMAL
2026-12-19 00:00:00	Predicted Traffic: 1597.37	Status: NORMAL
2026-12-20 00:00:00	Predicted Traffic: 1623.17	Status: NORMAL
2026-12-21 00:00:00	Predicted Traffic: 1634.31	Status: NORMAL
2026-12-22 00:00:00	Predicted Traffic: 1628.80	Status: NORMAL
2026-12-23 00:00:00	Predicted Traffic: 1612.09	Status: NORMAL
2026-12-24 00:00:00	Predicted Traffic: 1593.35	Status: NORMAL
2026-12-25 00:00:00	Predicted Traffic: 1580.96	Status: NORMAL
2026-12-26 00:00:00	Predicted Traffic: 1578.98	Status: NORMAL
2026-12-27 00:00:00	Predicted Traffic: 1586.34	Status: NORMAL
2026-12-28 00:00:00	Predicted Traffic: 1598.16	Status: NORMAL
2026-12-29 00:00:00	Predicted Traffic: 1608.59	Status: NORMAL
2026-12-30 00:00:00	Predicted Traffic: 1613.45	Status: NORMAL
2026-12-31 00:00:00	Predicted Traffic: 1611.70	Status: NORMAL

=====

FINAL RESULT

=====

No potential DDoS activity predicted.

=====

Forecast results saved to:
ddos_forecast_results.csv
/home/devan/meow/python/MCA.cpp/mlcs_exp_2.py:307: UserWarning: FigureCanvasAgg
is non-interactive, and thus cannot be shown
plt.show()
>

Experiment 3 output :

> /home/devan/meow/python/venv/bin/python
/home/devan/meow/python/MCA.cpp/mlcs_exp_3.py

===== DATASET INFORMATION =====

Dataset shape: (500, 9)

Column names:

['Duration', 'Protocol', 'Source_Bytes', 'Destination_Bytes', 'Packets', 'Flow_Rate',
'Source_Port', 'Destination_Port', 'Label']

First 5 records:

	Duration	Protocol	Source_Bytes	Destination_Bytes	Packets	Flow_Rate	Source_Port	Destination_Port	Label
0	4	UDP	7868	1025	507	15.13	28061	342	PortScan
1	29	TCP	1865	696	28	2.07	17803	80	Normal
2	1	UDP	6055	131	490	88.49	10393	383	PortScan
3	17	TCP	46737	1801	2446	88.81	56192	443	DDoS
4	5	UDP	93477	2200	7069	132.19	37610	80	DDoS

Target column: Label

Dataset after cleaning: (500, 9)

Encoding categorical features...

/home/devan/meow/python/MCA.cpp/mlcs_exp_3.py:114: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to `include` to select them, or to `exclude` to remove them and silence this warning.

See

https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

```
categorical_columns = X.select_dtypes(
```

Attack classes:

0 = BruteForce

1 = DDoS

2 = DoS

3 = Normal

4 = PortScan

Training samples: 400

Testing samples : 100

===== RANDOM FOREST =====

Training Random Forest...

Training completed!

===== MODEL RESULT =====

Random Forest Accuracy: 99.0 %

Classification Report:

	precision	recall	f1-score	support
BruteForce	1.00	1.00	1.00	20
DDoS	0.95	1.00	0.98	20
DoS	1.00	0.95	0.97	20
Normal	1.00	1.00	1.00	20
PortScan	1.00	1.00	1.00	20
accuracy		0.99	100	
macro avg	0.99	0.99	0.99	100
weighted avg	0.99	0.99	0.99	100

Confusion Matrix:

```
[[20 0 0 0 0]
 [ 0 20 0 0 0]
 [ 0 1 19 0 0]
 [ 0 0 0 20 0]
 [ 0 0 0 0 20]]
```

/home/devan/meow/python/MCA.cpp/mlcs_exp_3.py:248: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
plt.show()

===== FEATURE IMPORTANCE =====

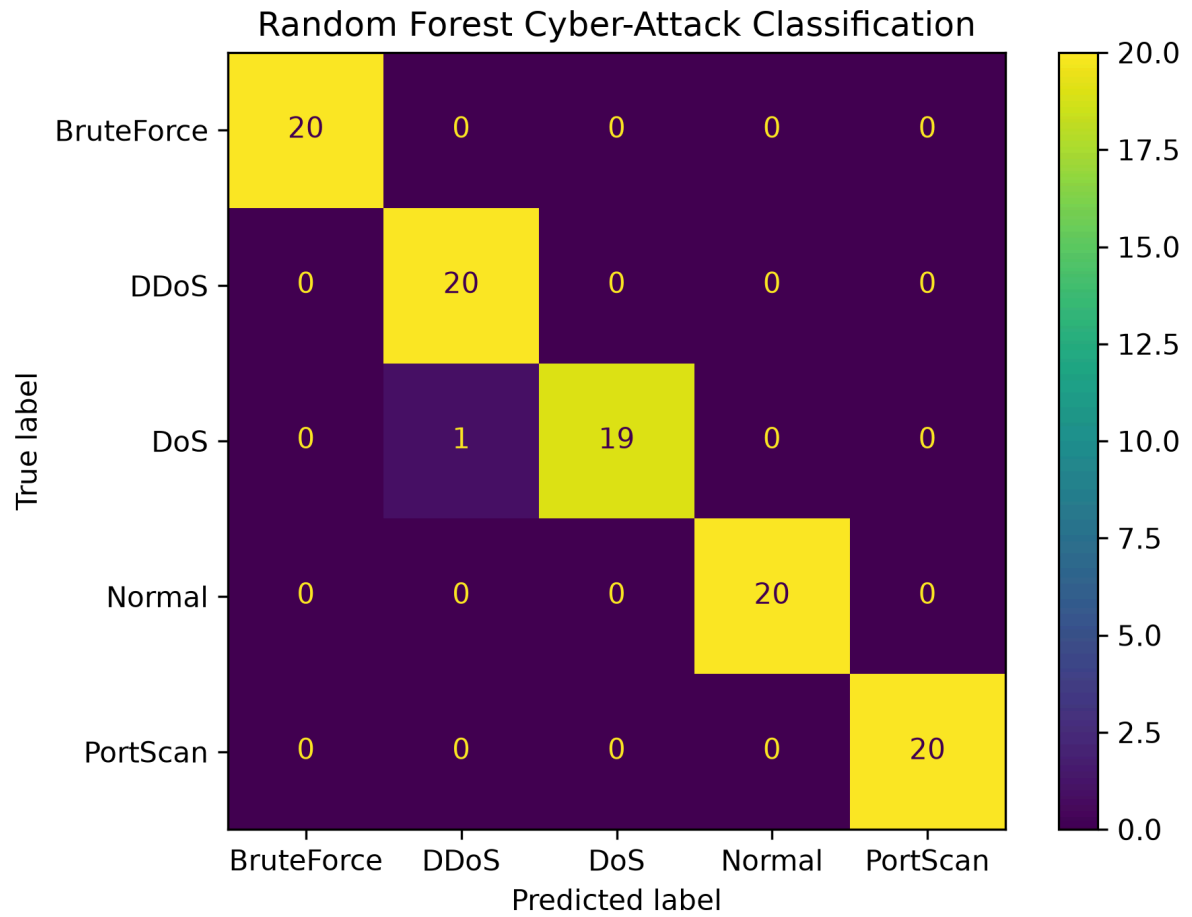
Feature Importance
Packets 0.220411
Source_Bytes 0.211768
Flow_Rate 0.208501
Destination_Port 0.117419
Duration 0.112488
Protocol 0.070342
Destination_Bytes 0.044199
Source_Port 0.014871

/home/devan/meow/python/MCA.cpp/mlcs_exp_3.py:299: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
plt.show()

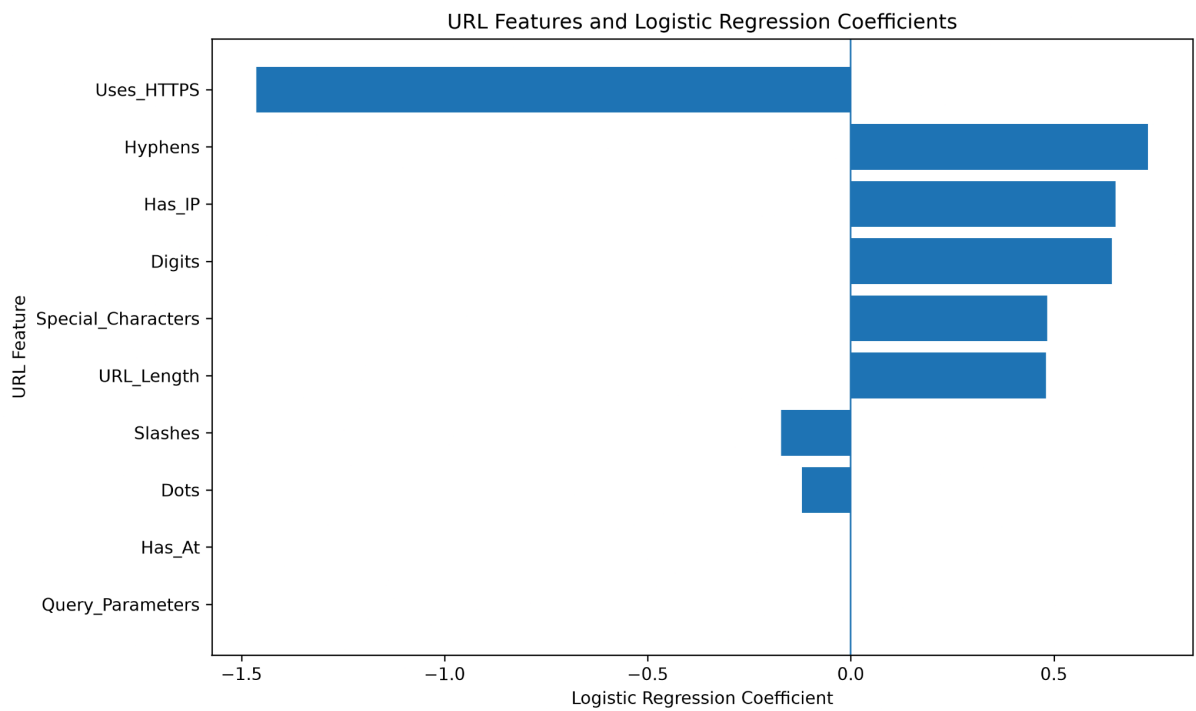
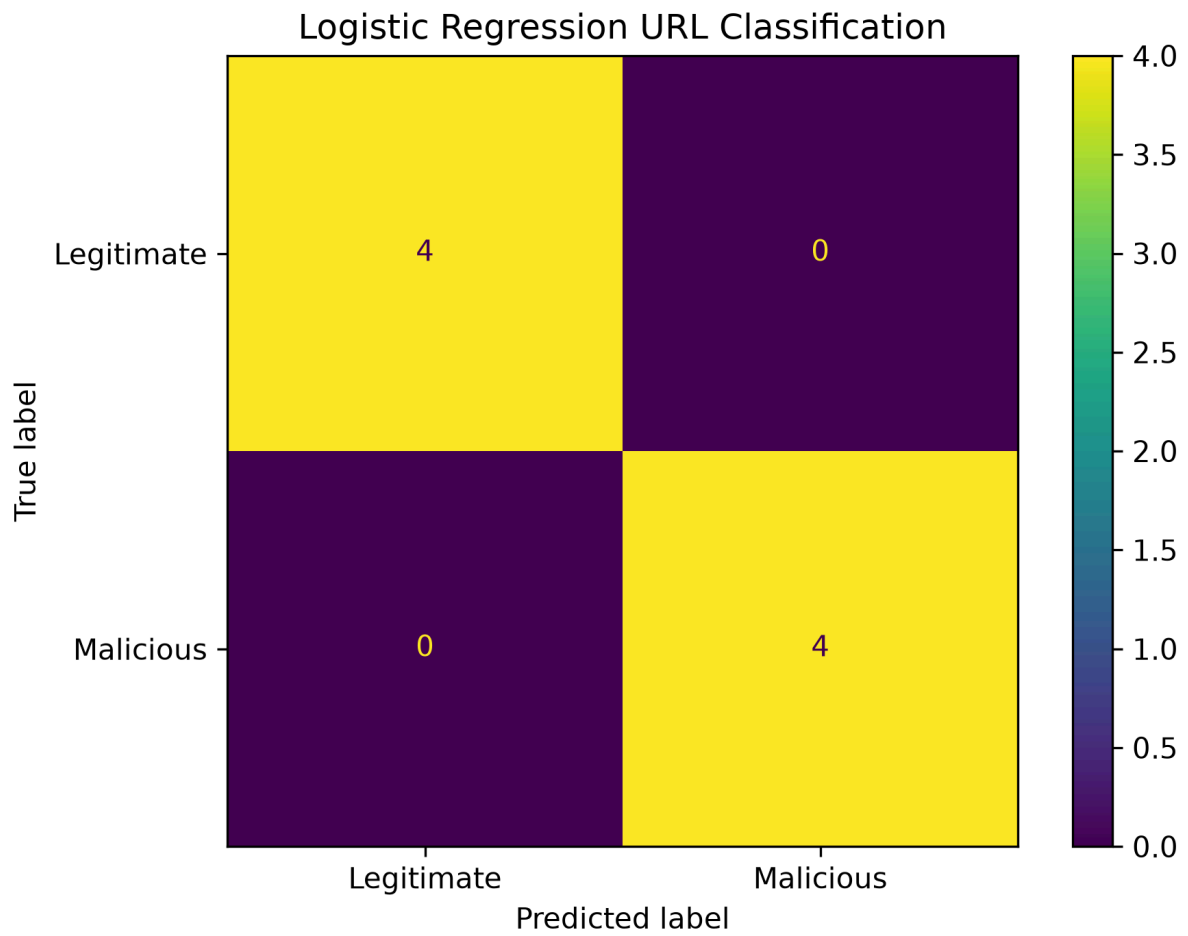
Predictions saved to: cyber_attack_predictions.csv

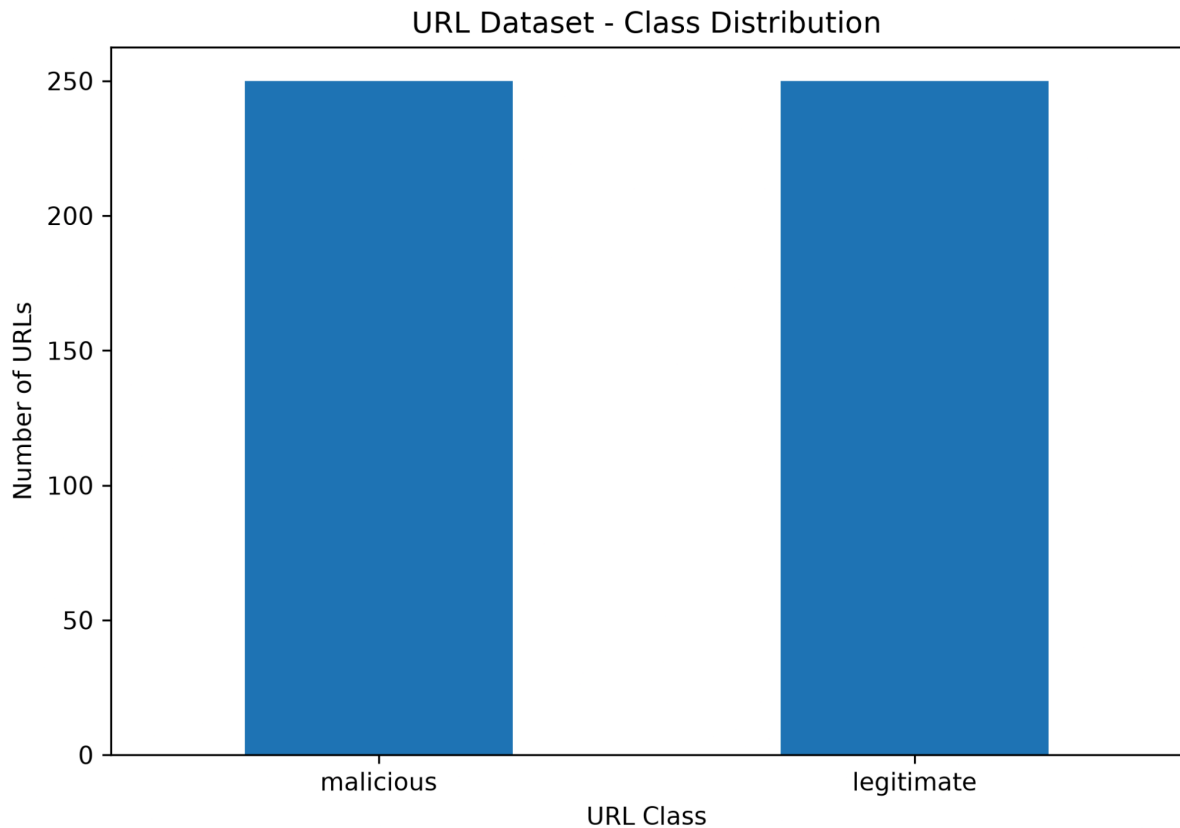
=====
FINAL RESULT
=====
Model: Random Forest Ensemble Classifier
Accuracy: 99.0 %
Number of trees: 100
=====

~/meow/python/MCA.cpp
✓ python



Experiment 4 output :





```
➤ /home/devan/meow/python/venv/bin/python  
/home/devan/meow/python/MCA.cpp/mlcs_exp_4.py
```

Dataset not found.
Creating a sample URL dataset...

Sample dataset created: urls.csv

===== DATASET INFORMATION =====

Dataset shape: (40, 2)

Column names:
['URL', 'Label']

First 10 records:

	URL	Label
0	https://www.google.com	legitimate
1	https://www.microsoft.com	legitimate
2	https://www.amazon.com	legitimate
3	https://www.wikipedia.org	legitimate
4	https://www.github.com	legitimate
5	https://www.linkedin.com	legitimate
6	https://www.apple.com	legitimate
7	https://www.ibm.com	legitimate

8 <https://www.python.org> legitimate
9 <https://www.amazon.in> legitimate

URL column: URL

Label column: Label

===== EXTRACTED FEATURES =====

	URL_Length	Dots	Hyphens	Slashes	Digits	Special_Characters	Has_At	Has_IP
0	22	2	0	2	0	5	0	0
1	25	2	0	2	0	5	0	0
2	22	2	0	2	0	5	0	0
3	25	2	0	2	0	5	0	0
4	22	2	0	2	0	5	0	0

Label distribution:

Label

0 20

1 20

Name: count, dtype: int64

Training samples: 32

Testing samples : 8

===== LOGISTIC REGRESSION =====

Training Logistic Regression...

Training completed!

===== MODEL RESULT =====

Logistic Regression Accuracy: 100.0 %

Classification Report:

	precision	recall	f1-score	support
Legitimate	1.00	1.00	1.00	4
Malicious	1.00	1.00	1.00	4
accuracy			1.00	8
macro avg	1.00	1.00	1.00	8
weighted avg	1.00	1.00	1.00	8

Confusion Matrix:

```
[[4 0]
 [0 4]]
```

Confusion matrix saved as: url_confusion_matrix.png

===== FEATURE COEFFICIENTS =====

	Feature Coefficient	Absolute_Coefficient
Uses_HTTPS	-1.463558	1.463558
Hyphens	0.732242	0.732242
Has_IP	0.651244	0.651244
Digits	0.643141	0.643141
Special_Characters	0.483876	0.483876
URL_Length	0.480553	0.480553
Slashes	-0.171695	0.171695
Dots	-0.120806	0.120806
Has_At	0.000000	0.000000
Query_Parameters	0.000000	0.000000

Feature coefficient graph saved as: url_feature_coefficients.png

Predictions saved to: url_predictions.csv

=====

FINAL RESULT

=====

Model: Logistic Regression
Task: URL Classification
Classes: Legitimate / Malicious
Accuracy: 100.0 %

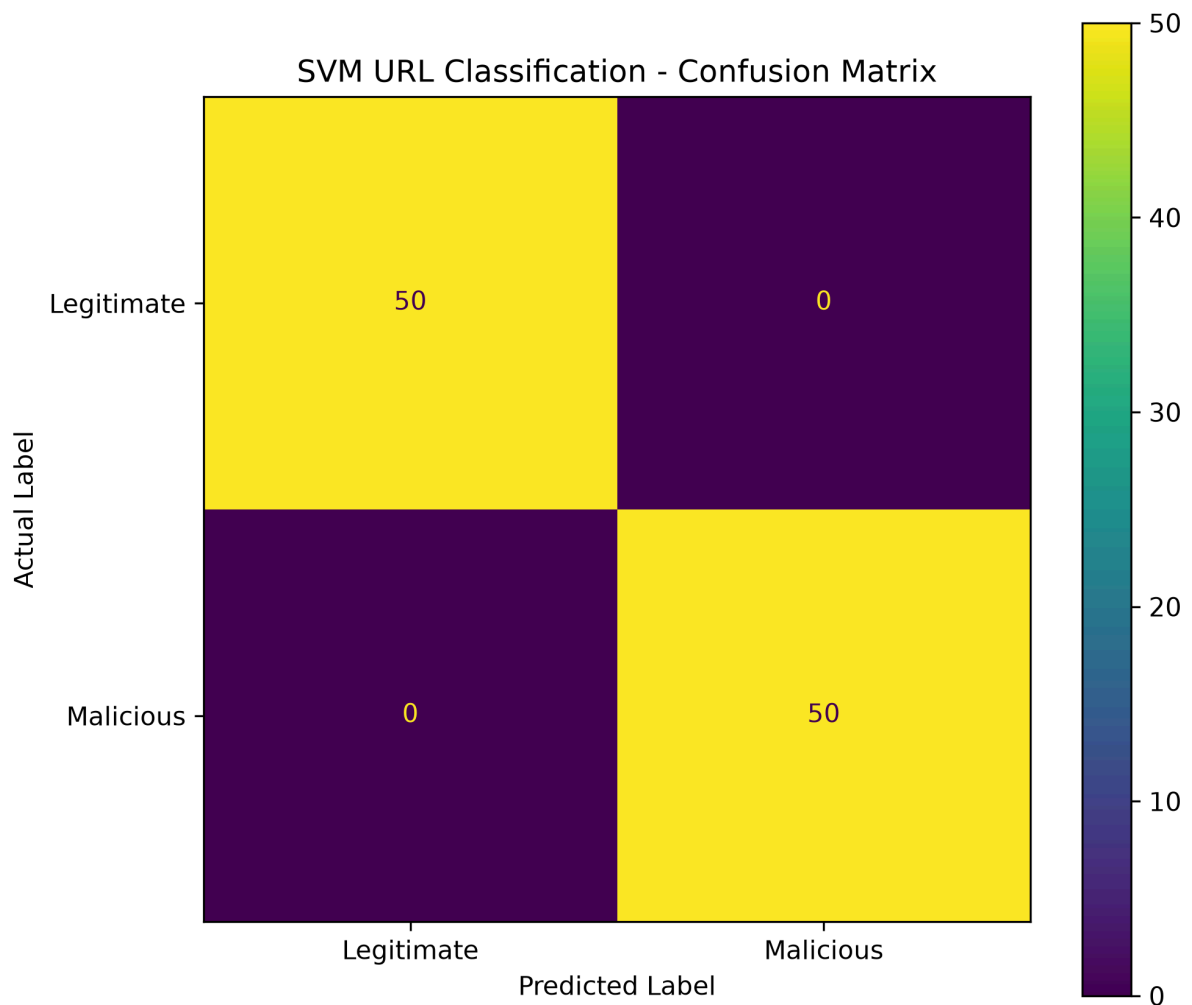
=====

Files generated:

1. urls.csv
2. url_confusion_matrix.png
3. url_feature_coefficients.png
4. url_predictions.csv

Program completed successfully!
~/meow/python/MCA.cpp

Experiment 5 output :



```
> /home/devan/meow/python/venv/bin/python  
/home/devan/meow/python/MCA.cpp/mlcs_exp_5.py
```

```
=====
```

SVM URL CLASSIFICATION

```
=====
```

Dataset path:

/home/devan/meow/python/MCA.cpp/url_svm_dataset.csv

Dataset successfully loaded!

```
=====
```

DATASET INFORMATION

```
=====
```

Dataset Shape:

(500, 2)

Column Names:

['URL', 'Label']

First 10 Rows:

	URL	Label
0	http://amazon-confirm.click/secure/update-pass...	malicious
1	http://update-account.site/update-billing.php	malicious
2	https://www.github.com/products	legitimate
3	https://www.netflix.com/login	legitimate
4	https://www.amazon.com/products	legitimate
5	http://wallet-connect.click/wallet/connect	malicious
6	http://account-check.site/urgent-security-chec...	malicious
7	https://www.python.org/products	legitimate
8	https://www.ibm.com/about	legitimate
9	https://www.linkedin.com/account	legitimate

Missing Values:

URL 0

Label 0

dtype: int64

CLASS DISTRIBUTION

Label

malicious 250

legitimate 250

Name: count, dtype: int64

Class distribution graph saved:

/home/devan/meow/python/MCA.cpp/url_class_distribution.png

LABEL ENCODING

0 = Legitimate

1 = Malicious

Total URLs:

500

Legitimate URLs:

250

Malicious URLs:

250

TRAIN TEST SPLIT

Training samples:

400

Testing samples:

100

TF-IDF FEATURE EXTRACTION

Training TF-IDF shape:

(400, 2084)

Testing TF-IDF shape:

(100, 2084)

SVM MODEL TRAINING

SVM training completed successfully!

MODEL PREDICTION

MODEL PERFORMANCE

Accuracy : 100.00%

Precision: 100.00%

Recall : 100.00%

F1-Score : 100.00%

CLASSIFICATION REPORT

	precision	recall	f1-score	support
Legitimate	1.00	1.00	1.00	50
Malicious	1.00	1.00	1.00	50
accuracy			1.00	100
macro avg	1.00	1.00	1.00	100

weighted avg 1.00 1.00 1.00 100

=====

CONFUSION MATRIX

=====

Confusion Matrix:

```
[[50  0]
 [ 0 50]]
```

Confusion matrix graph saved:

/home/devan/meow/python/MCA.cpp/svm_confusion_matrix.png

=====

CUSTOM URL PREDICTION

=====

URL:

<https://www.google.com>

Prediction:

LEGITIMATE

URL:

<https://www.microsoft.com/login>

Prediction:

LEGITIMATE

URL:

<https://www.amazon.com/products>

Prediction:

LEGITIMATE

URL:

<https://www.github.com>

Prediction:

LEGITIMATE

URL:

<https://www.python.org/docs>

Prediction:

LEGITIMATE

URL:

<http://paypal-security.xyz/login/verify-account>

Prediction:

MALICIOUS

URL:
<http://free-prize.xyz/login.php>
Prediction:
MALICIOUS

URL:
<http://google-verify.tk/secure/update-password>
Prediction:
MALICIOUS

URL:
<http://bank-security.top/account/confirm>
Prediction:
MALICIOUS

URL:
<http://wallet-connect.click/wallet/connect>
Prediction:
MALICIOUS

Custom URL predictions saved:
/home/devan/meow/python/MCA.cpp/svm_url_predictions.csv

=====

FINAL PROJECT SUMMARY

=====

Dataset Size : 500 URLs

Training Samples : 400

Testing Samples : 100

Legitimate URLs : 250

Malicious URLs : 250

Algorithm : Support Vector Machine (SVM)

Kernel : Linear

C Parameter : 1.0

Feature Extraction : Character-level TF-IDF

N-Gram Range : 2 - 5

Accuracy : 100.00%

Precision : 100.00%

Recall : 100.00%

F1-Score : 100.00%

=====

OUTPUT FILES

=====

1. url_class_distribution.png
2. svm_confusion_matrix.png
3. svm_url_predictions.csv

=====

PROGRAM COMPLETED SUCCESSFULLY

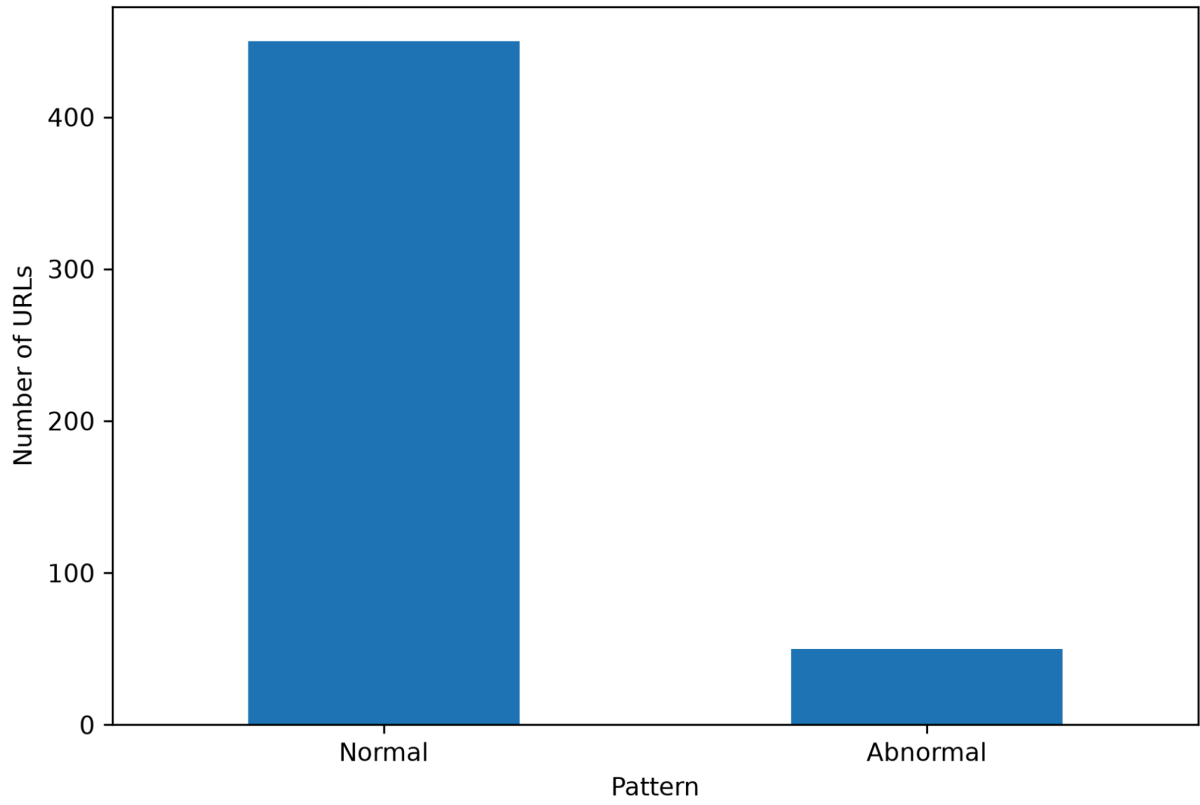
=====

~/meow/python

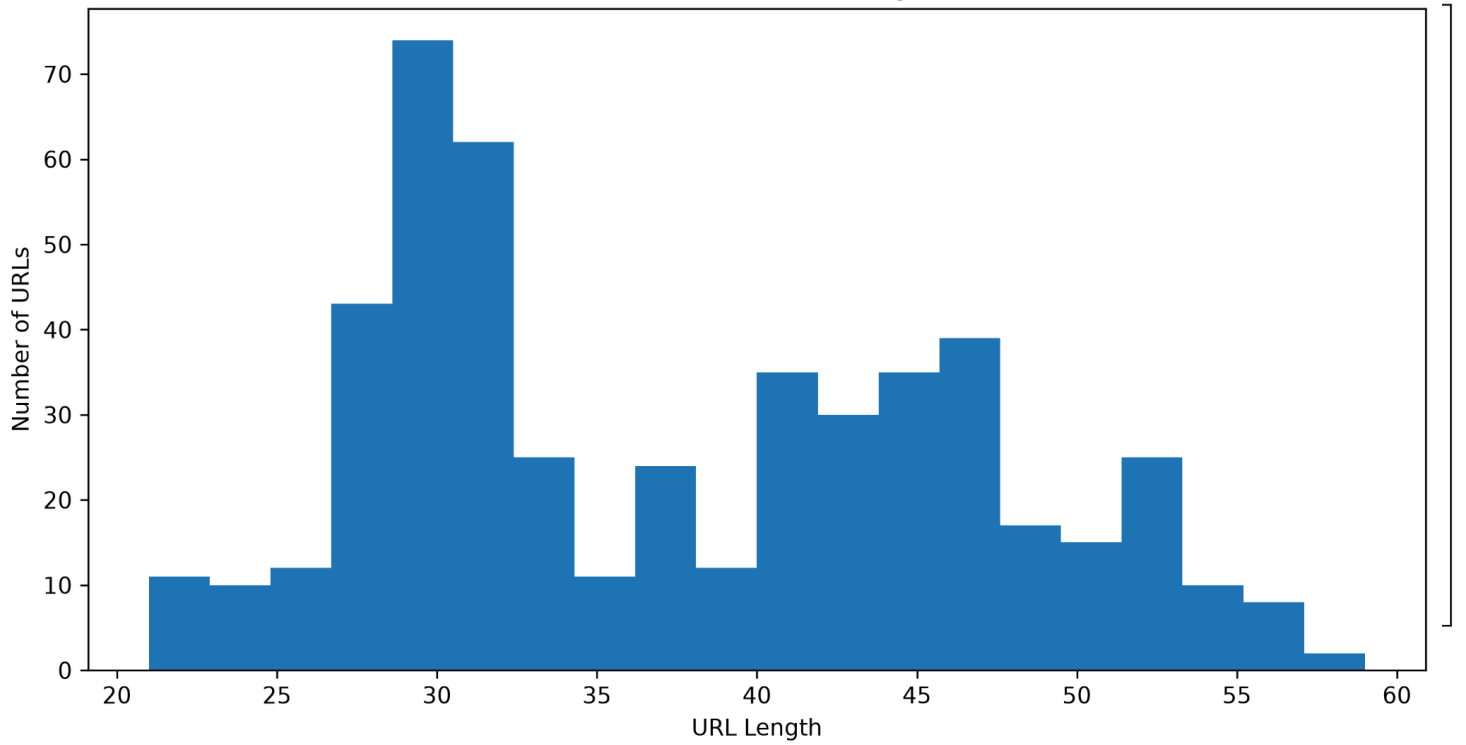
✓ python

Experiment 6 output :

Normal vs Abnormal URL Patterns



Distribution of URL Length



```
➤ /home/devan/meow/python/venv/bin/python
/home/devan/meow/python/MCA.cpp/mlcs_exp_6.py
```

```
=====
URL ABNORMAL PATTERN DETECTION
USING FEATURE ENGINEERING
=====
```

```
Dataset path:
/home/devan/meow/python/MCA.cpp/url_svm_dataset.csv
```

```
Dataset successfully loaded!
```

```
=====
DATASET INFORMATION
=====
```

```
Dataset Shape:
(500, 2)
```

```
Columns:
['URL', 'Label']
```

```
First 5 URLs:
```

	URL	Label
0	http://amazon-confirm.click/secure/update-pass...	malicious
1	http://update-account.site/update-billing.php	malicious
2	https://www.github.com/products	legitimate
3	https://www.netflix.com/login	legitimate
4	https://www.amazon.com/products	legitimate

```
=====
EXTRACTING URL FEATURES
=====
```

```
Extracted Features:
['URL_Length', 'Hostname_Length', 'Dot_Count', 'Slash_Count', 'Hyphen_Count',
'Underscore_Count', 'Question_Count', 'Equal_Count', 'At_Count', 'Ampersand_Count',
'Percent_Count', 'Digit_Count', 'Letter_Count', 'Digit_Ratio', 'Special_Character_Count',
'HTTPS', 'Has_At_Symbol', 'Has_Double_Slash', 'Has_IP_Address',
'Suspicious_Keyword_Count', 'Suspicious_TLD', 'URL_Entropy']
```

```
Feature Dataset Shape:
(500, 22)
```

```
=====
FEATURE VALUES
=====
```

URL_Length Hostname_Length Dot_Count Slash_Count Hyphen_Count
 Underscore_Count Question_Count Equal_Count At_Count Ampersand_Count
 Percent_Count Digit_Count Letter_Count Digit_Ratio Special_Character_Count HTTPS
 Has_At_Symbol Has_Double_Slash Has_IP_Address Suspicious_Keyword_Count
 Suspicious_TLD URL_Entropy

	50	20	1	4	2	0	0	0	0
0	0	0	42	0.0		8	0	0	0
0		4	1	4.353270					
	45	19	2	3	2	0	0	0	0
0	0	0	37	0.0		8	0	0	0
0		3	1	4.049767					
	31	14	2	3	0	0	0	0	0
0	0	0	25	0.0		6	1	0	0
0		0	0	3.937752					
	29	15	2	3	0	0	0	0	0
0	0	0	23	0.0		6	1	0	0
0		1	0	4.021268					
	31	14	2	3	0	0	0	0	0
0	0	0	25	0.0		6	1	0	0
0		0	0	3.953566					
	42	20	1	4	1	0	0	0	0
0	0	0	35	0.0		7	0	0	0
0		1	1	3.663051					
	51	18	2	3	3	0	0	0	0
0	0	0	42	0.0		9	0	0	0
0		3	1	3.990260					
	31	14	2	3	0	0	0	0	0
0	0	0	25	0.0		6	1	0	0
0		0	0	3.824533					
	25	11	2	3	0	0	0	0	0
0	0	0	19	0.0		6	1	0	0
0		0	0	3.753270					
	32	16	2	3	0	0	0	0	0
0	0	0	26	0.0		6	1	0	0
0		1	0	4.069549					

===== ANOMALY DETECTION =====

Normal URLs:

450

Abnormal URLs:

50

===== ABNORMAL URL PATTERNS =====

```

=====
URL      Label Pattern
http://wallet-connect.click/wallet/connect  malicious Abnormal
http://account-verify.net/verify@free-prize.xyz/login  malicious Abnormal
http://account-verify.net/verify@bank-security.top/login  malicious Abnormal
http://user-check.info/verify@verify-user.online/login  malicious Abnormal
http://free-prize.xyz/urgent-security-check.php  malicious Abnormal
http://user-check.info/verify@secure-account.xyz/login  malicious Abnormal
http://user-check.info/verify@secure-account.xyz/login  malicious Abnormal
http://account-verify.net/verify@wallet-connect.click/login  malicious Abnormal
http://user-check.info/verify@amazon-confirm.click/login  malicious Abnormal
http://user-check.info/verify@google-verify.tk/login  malicious Abnormal
http://secure-login.com/verify@wallet-connect.click/login  malicious Abnormal
http://free-prize.xyz/urgent-security-check.php  malicious Abnormal
http://user-check.info/verify@verify-user.online/login  malicious Abnormal
http://account-verify.net/verify@bank-security.top/login  malicious Abnormal
https://www.stackoverflow.com/dashboard  legitimate Abnormal
https://www.apple.com  legitimate Abnormal
https://www.stackoverflow.com  legitimate Abnormal
http://free-prize.xyz/secure/update-password  malicious Abnormal
http://user-check.info/verify@google-verify.tk/login  malicious Abnormal
http://user-check.info/verify@secure-account.xyz/login  malicious Abnormal
https://www.adobe.com  legitimate Abnormal
https://www.microsoft.com  legitimate Abnormal
http://secure-login.com/verify@wallet-connect.click/login  malicious Abnormal
https://www.microsoft.com  legitimate Abnormal
https://www.python.org  legitimate Abnormal
http://account-verify.net/verify@verify-user.online/login  malicious Abnormal
http://free-prize.xyz/free-prize/login.php  malicious Abnormal
http://user-check.info/verify@paypal-security.xyz/login  malicious Abnormal
http://free-prize.xyz/signin/verify  malicious Abnormal
http://user-check.info/verify@login-alert.info/login  malicious Abnormal
http://account-verify.net/verify@paypal-security.xyz/login  malicious Abnormal
http://account-verify.net/verify@login-alert.info/login  malicious Abnormal
http://user-check.info/verify@login-alert.info/login  malicious Abnormal
https://www.wikipedia.org  legitimate Abnormal
http://user-check.info/verify@secure-account.xyz/login  malicious Abnormal
http://user-check.info/verify@bank-security.top/login  malicious Abnormal
http://account-verify.net/verify@secure-account.xyz/login  malicious Abnormal
https://www.stackoverflow.com  legitimate Abnormal
http://wallet-connect.click/wallet/connect  malicious Abnormal
https://www.python.org  legitimate Abnormal
http://user-check.info/verify@secure-account.xyz/login  malicious Abnormal
https://www.facebook.com  legitimate Abnormal
http://secure-login.com/verify@login-alert.info/login  malicious Abnormal
http://secure-login.com/verify@google-verify.tk/login  malicious Abnormal
http://account-verify.net/verify@secure-account.xyz/login  malicious Abnormal
https://www.apple.com  legitimate Abnormal

```

http://user-check.info/verify@verify-user.online/login malicious Abnormal
http://user-check.info/verify@bank-security.top/login malicious Abnormal
https://www.apple.com legitimate Abnormal
https://www.stackoverflow.com/login legitimate Abnormal

Feature dataset saved:

/home/devan/meow/python/MCA.cpp/url_engineered_features.csv

Anomaly graph saved:

/home/devan/meow/python/MCA.cpp/url_anomaly_distribution.png

URL length graph saved:

/home/devan/meow/python/MCA.cpp/url_length_distribution.png

Suspicious keyword graph saved:

/home/devan/meow/python/MCA.cpp/suspicious_keyword_distribution.png

=====

FINAL SUMMARY

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Total URLs : 500

Engineered Features : 22

Normal URLs : 450

Abnormal URLs : 50

Detection Algorithm : Isolation Forest

Feature Engineering : URL length, character counts,
digit ratio, HTTPS, IP address,
suspicious keywords, suspicious TLD,
URL entropy and special characters

=====

OUTPUT FILES

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1. url_engineered_features.csv
2. url_anomaly_distribution.png
3. url_length_distribution.png
4. suspicious_keyword_distribution.png

=====

PROGRAM COMPLETED SUCCESSFULLY

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~/meow/python
✓ python